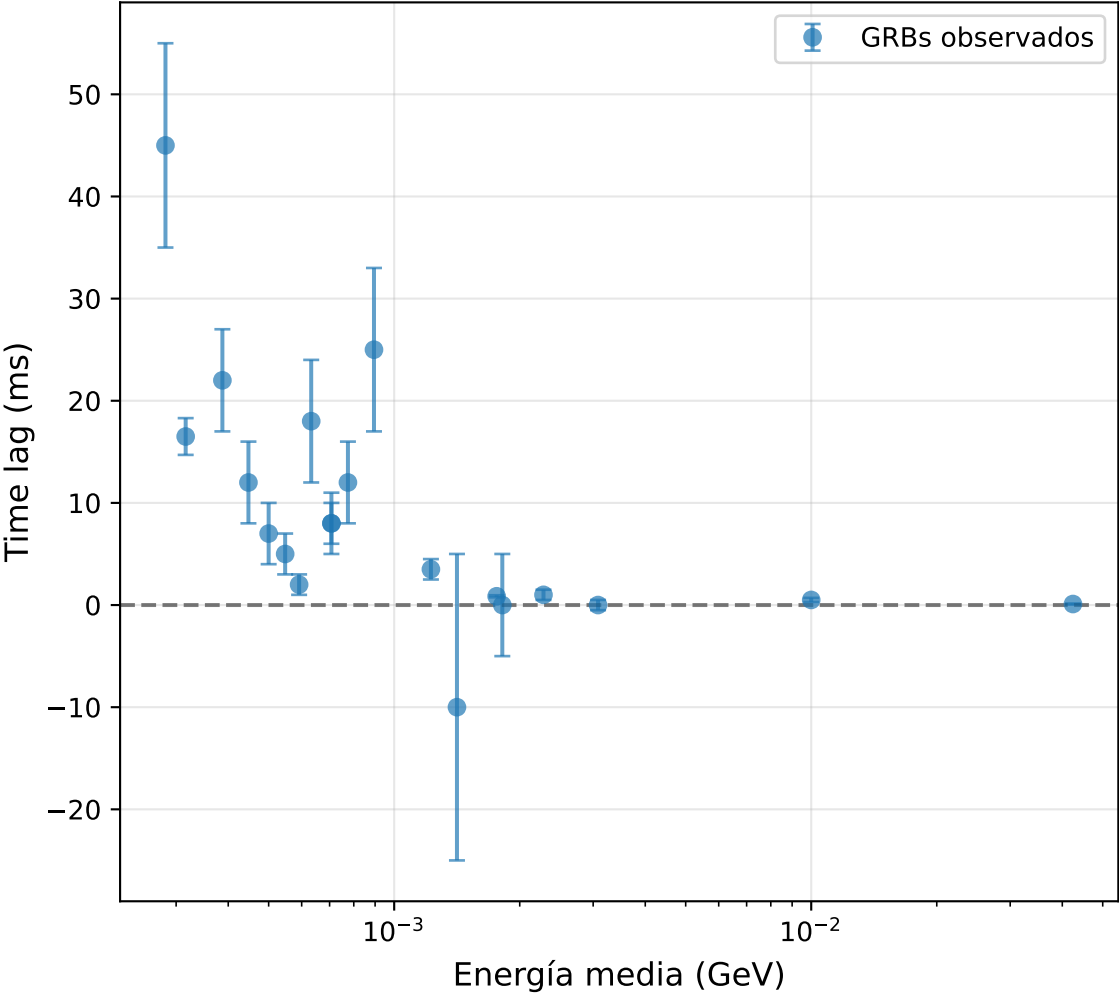
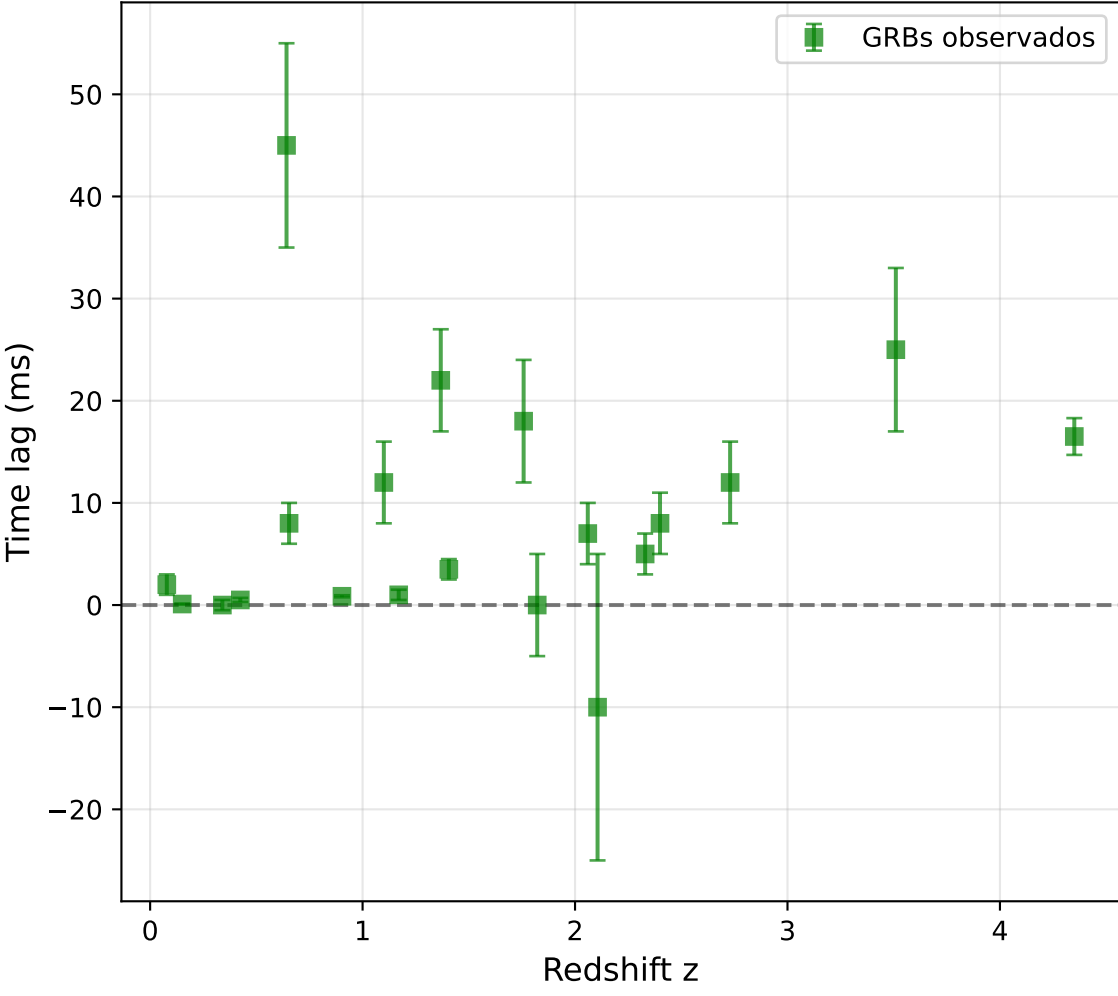


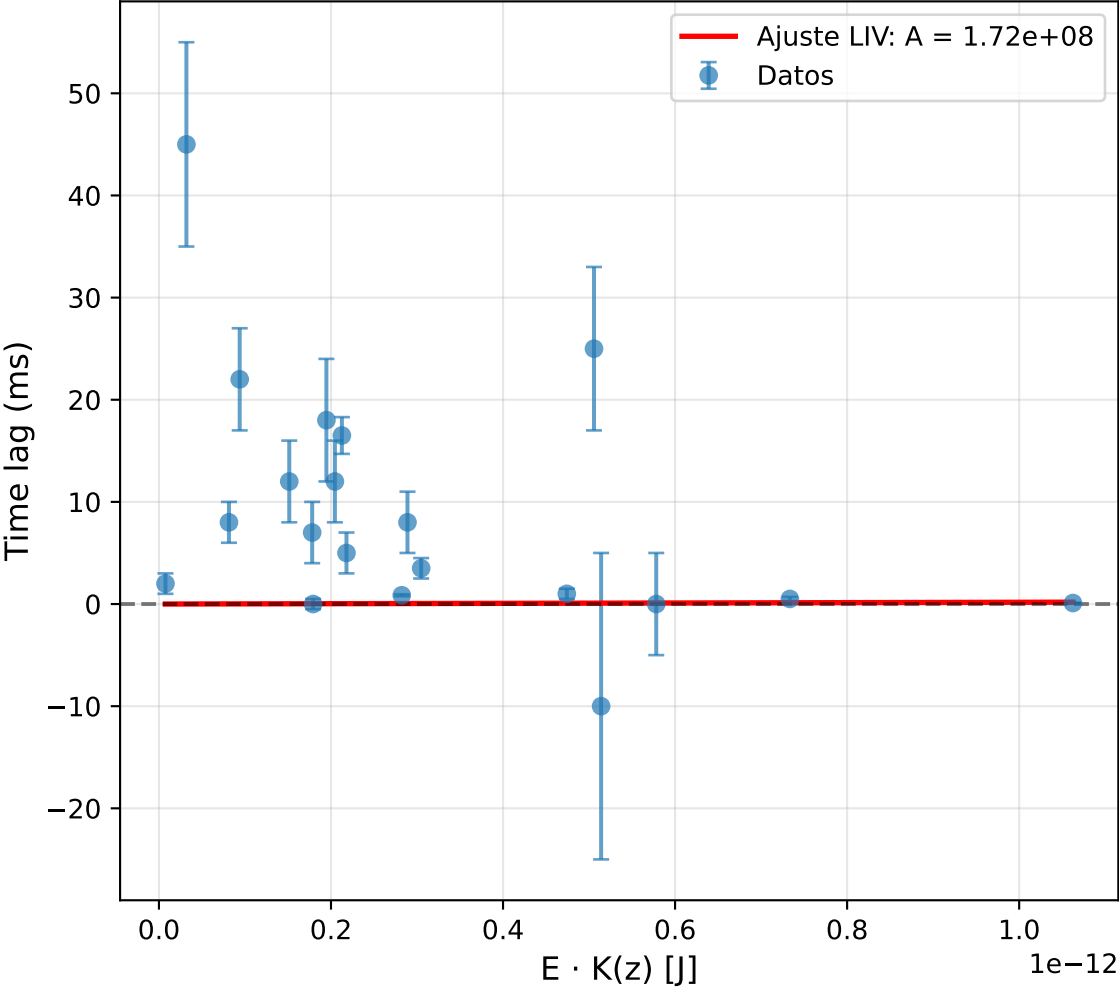
Lag vs Energía
Spearman $\rho = -0.747$, $p = 0.000$



Lag vs Redshift
Spearman $\rho = 0.467$, $p = 0.038$



Ajuste modelo LIV
 $\chi^2/\text{dof} = 15.05$, $E_{\text{QG}}/E_{\text{Pl}} = 6.8e-01$



RESULTADOS TEST #5: VSL EN GRBs

MUESTRA: 20 GRBs con lag espectral medido
Rango de z: 0.079 - 4.35
Rango de E: 800 keV - 18 GeV

CORRELACIONES

Lag vs Energia: $\rho = -0.747$, $p = 0.0002$
Lag vs Redshift: $\rho = 0.467$, $p = 0.0377$

TEST DE HIPOTESIS NULA

$\langle \text{lag} \rangle = 0.30 \pm 0.04$ ms
Z-score: 6.98
P-value: 0.000000

AJUSTE MODELO LIV

$E_{\text{QG}} / E_{\text{Planck}} > 6.8e-01$
 $\chi^2/\text{dof} = 15.05$

CONCLUSION: LAG INTRÍNSECO (correlación negativa = efecto de fuente)

INTERPRETACION OCTH:
Posible evidencia de $c_{\text{eff}} = c \cdot \Psi(E)$